Teerth Patel

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Education

University of Texas at Austin **May 2028**

Bachelor of Science in Electrical and Computer Engineering; GPA: 3.75 / 4.0 *Austin, TX*

- Relevant Coursework: Digital Logic Design, VLSI Design, Embedded Systems, Circuit Theory, Signals and Systems

Technical Skills

Languages: SystemVerilog, Verilog, C, C++, Python, Perl, Assembly, Tcl
RTL / Digital: RTL Design, FPGA, FSMs, Digital Logic, Clock-Domain Crossing
EDA / Engineering: Cadence Virtuoso, Synopsys PrimeTime, HSPICE, Vivado, KiCad, LTspice, Altium, MATLAB
Interfaces / Validation: SPI, UART, RS-485, GPIO, PWM, Oscilloscopes, Logic Analyzers
Development: Linux

Experience

Rice University **May 2026 – October 2026**

Undergraduate Researcher, ECE

- Integrate TI AWR1843 and Analog Devices TinyRad radar with ZED stereo vision on an NVIDIA Jetson Xavier NX, coordinating embedded digital sensor interfaces for synchronized acquisition.
- Develop Python/Linux acquisition routines and validate 300+ synchronized frames through timestamp and structural checks, debugging communication across sensing and processing subsystems.

Corveus Medical **May 2026 – September 2026**

Electrical Engineering Intern

- Design a mixed-signal waveform generator around a DAC8831 16-bit DAC, SPI control, isolated $\pm$15 V power, precision signal conditioning, and a Howland current-source output stage.
- Develop Python/FTDI control and validation software for 16-bit command generation, device initialization, waveform updates, and oscilloscope-based hardware bring-up.

PSMS Controls Services **Jan 2026 – September 2026**

Hardware Engineering Intern

- Debug UART, SPI, and RS-485 interfaces with oscilloscopes and logic analyzers, resolving 20+ timing, grounding, and signal-integrity faults during system commissioning.
- Validate 50+ analog and digital I/O channels while analyzing interface behavior, power distribution, filtering, and transient protection across embedded-control hardware.

Projects

Asynchronous FIFO & CDC Verification | *SystemVerilog, Vivado* **Aug 2026 – Present**

- Designing a dual-clock asynchronous FIFO in SystemVerilog using Gray-coded read/write pointers and two-flop synchronizers for safe clock-domain crossing.
- Developing a self-checking verification environment with assertions, randomized stimulus, and functional coverage to validate data integrity, pointer synchronization, and full/empty behavior.

FPGA RTL Stopwatch | *Verilog, Vivado, Xilinx Basys 3 / Artix-7* **Feb 2026 – Apr 2026**

- Designed modular Verilog RTL with finite-state control, synchronous counters, clock division, input debouncing, and seven-segment display logic.
- Implemented and validated the design in Vivado on an Artix-7 FPGA with hardware-controlled start, stop, reset, and timing functions.

Technical Leadership & Organizations

Longhorn Neurotech | *Hardware Systems Team* **Sep 2025 – Present**

- Integrate LiDAR, RGB, ultrasonic, and motor-control interfaces across embedded hardware systems.
- Coordinate hardware/software timing to maintain $\leq$5 ms synchronization within a 500 Hz closed-loop architecture.

Longhorn Baja | *Research & Development – Electronics Lead* **Aug 2025 – Present**

- Lead validation across 15+ vehicle electrical subsystems, diagnosing communication, grounding, connector, and voltage faults.
- Standardize electrical debugging and validation workflows, contributing to 30% greater vehicle uptime.